



# Build Production Ready AI-powered Search Application

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# What we'll cover today

- Quick Intro on OpenSearch
- Search and Vector Concepts
- Vectors 101 – Notion of Vectors in Search
- Semantic Search – what is KNN?
- How does AI Powered Search work?
- OpenSearch Serverless – Vector Database

# HOW AI IS CHANGING SEARCH:

## The Paradigm Shift



### TRADITIONAL SEARCH (Keywords)

Q best camera for hiking 2024



1. Keyword Matching  
(Best, Camera, Hiking)



2. Lists of  
Indexed Links



3. Requires  
Manual Review



4. Static Results



### AI-POWERED DISCOVERY (Intent & Context)

I need a compact, weather-sealed camera for a 3-day mountain trek, budget \$1200.



1. Natural Language  
Understanding



2. Generative Answers  
& Summaries



3. Personalized  
Recommendations



4. Iterative  
Dialogue



OpenSearch is a **community-driven, open-source** search, analytics, and vector database platform with integrated tools for observability, security, visualization, and AI-powered applications

Roadmap:



# OpenSearch by the numbers

**1.4B+**

Project downloads

**1M+**

Monthly page views  
For [opensearch.org](https://opensearch.org)

**100+**

Solution providers

**3K+**

Active contributors

**400+**

Active organizations

In the top

**20**

Of LF projects by  
contributor activity

**130+**

GitHub repositories

**4K+**

Slack workspace  
members

**7K+**

User forum members

# Amazon OpenSearch Service

Simplify AI-powered search, observability, and vector database operations with a secure, cost-effective managed service



## Operational Simplicity

Fully managed OpenSearch in the AWS Cloud. Serverless and managed clusters. API-driven deployments, upgrades, and patches



## Performance

6x performance gains from OpenSearch 1.3 to current version. Single-digit ms latencies for lexical and vector queries



## Cost efficiency

Storage tiering for voluminous log data, specialized instances, tune cluster sizes or autoscaled to match request traffic



## Integrations

Quickly and easily connect all of your data for faster, better insights

# Amazon OpenSearch Service features

DEPLOY FAST, AT SCALE, WITH REMARKABLE VALUE



## Automated

API-driven deployment,  
and reconfiguration

Service-managed  
upgrades and patches

Serverless Autoscale

24x7 monitoring, self-  
healing

Cross-cluster search



## Cost-optimized

Optimize infrastructure,  
optimize cost

Reserved instances Up to  
52% discount

Tiered storage,  
specialized instances

Service snapshots  
included

Cross-zone transfer  
included



## Resilient

One-click, multi-zone  
deployments, up to  
99.99% SLA

Automatic hourly  
snapshots, 2-week  
retention

Serverless enhanced  
resilience with S3

Specialized instance  
types using S3 as a  
backing store



## Secure

Fine-grained access  
control

Encryption at rest and in  
flight, customer-  
managed keys

Audit logging

SAML/Cognito/IAM  
integration



## Integrated

Integrated user interface

One-click integration  
with DynamoDB and  
Document DB

Zero ETL access to logs in  
S3 and CloudWatch Logs

Supports OpenTelemetry  
and Jaeger



FROM KEYWORDS TO UNDERSTANDING

# Search and Vector Concepts

Search is all about *retrieving relevant information* in response to a user goal





# The Evolution of Search

FROM KEYWORDS TO UNDERSTANDING



Keyword

Semantic

Hybrid

Agentic

Search is all about *retrieving relevant information* in response to a user goal

# Lexical Search

- Search engines match terms to objects in a catalog
- Match free text or structured data
- Relevance determines sort order
- Facets enable users to drill in to the search results to narrow by attribute value

amazon prime

All 8 foot blue couch

All Clinic Prime Video Groceries Pharmacy Buy Again Amazon Business Livestreams Health & Household Products

1-48 of 109 results for "8 foot blue couch"

**Amazon Prime**  
☐ prime

**Delivery Day**  
☐ Get It by Tomorrow

**More-sustainable Products**  
☐ Climate Pledge Friendly

**Department**  
Sofas & Couches  
Living Room Furniture Sets  
Futons  
Futon Sets

**Customer Reviews**  
★★★★☆ & Up  
★★★★☆ & Up  
★★★★☆ & Up  
★★★★☆ & Up

**Brands**  
☐ Stone & Beam  
☐ Rivet  
☐ Christopher Knight Home  
☐ Signature Design by Ashley  
☐ Best Choice Products  
☐ HONBAY  
☐ Modway

**Furniture Price**  
Under \$100  
\$100 to \$500  
\$500 to \$1,000  
\$1,000 to \$2,500  
\$ Min \$ Max Go

**Deals & Discounts**  
All Discounts  
Today's Deals

**Sofa Type**  
☐ Convertible  
☐ Loveseat  
☐ Sectional  
☐ Standard

**Furniture & Décor Style**  
☐ Bohemian

**ORRD** Add a warm sofa to your home  
Shop ORRD >

**Chesterfield Sofa, Modern Tufted Velvet Couch 3 Seater with Scroll Arms and Nailhead for Living Ro...**  
★★★★☆ 55

**Chesterfield Sofa for Living Room, 3 Seater Sofa Couch Faux Leather Fabric Home Seating Couch S...**  
★★★★☆ 81

**Results**  
Price and other details may vary based on product size and color.

**HONBAY Convertible Sectional Sofa U Shaped Couch with Ottomans,...**  
★★★★☆ ~ 1,980  
**\$1,129<sup>99</sup>** Was: \$1,199.99  
FREE delivery Jun 20 - 21  
Options: 4 sizes

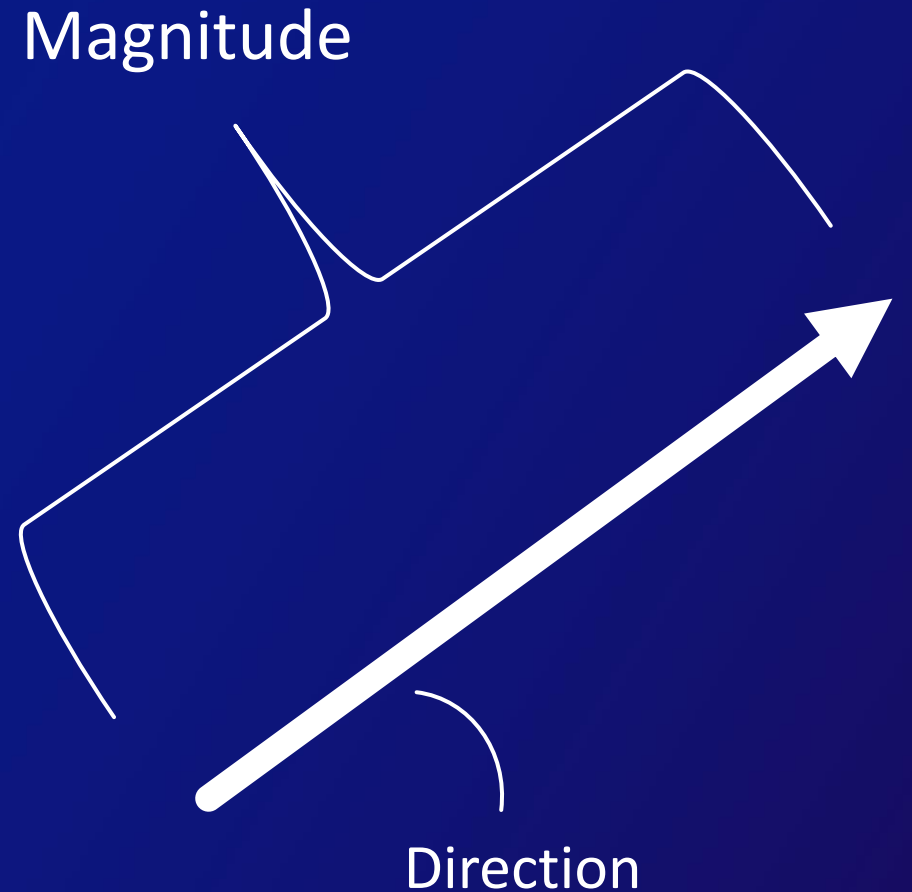
**Amazon Brand – Rivet Frederick Mid-Century Channel Tufted Velvet Sofa...**  
★★★★☆ ~ 149  
**\$1,144<sup>65</sup>**  
prime  
FREE delivery as soon as Mon, Jun 19, 8 AM - 8 PM  
Only 10 left in stock - order soon.  
Options: 2 sizes  
Available Services: Deluxe Delivery and Unpack and Room of Choice Delivery  
Amazon brand

# Vectors 101

“I am attending a great webinar about vector search “

===

[0.743, 0.720, -0.325, 0.195, 0.835, -0.945]



# Notion of vectors in search

A *vector* is a list of values describing some attributes of an item.

$$v_1 = [5, 4, 854, \$1.1M]$$



1. How many bedrooms?

2. How many bathrooms?

3. Size of the house in sqmeters?

$$v_3 = [6, 4, 530, \$2.1M]$$



4. Price of the house?



$$v_2 = [4, 3, 335, \$530K]$$



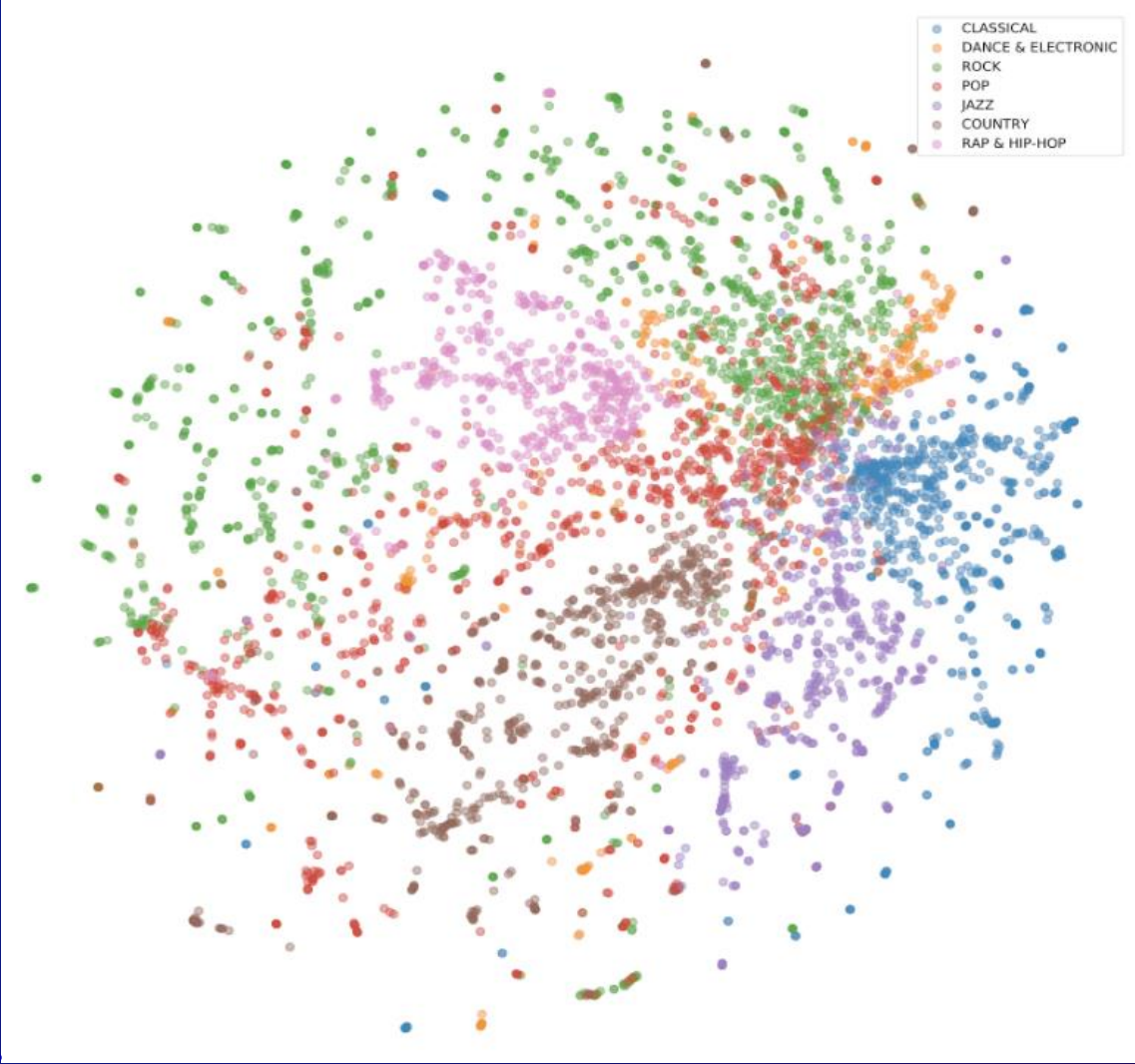
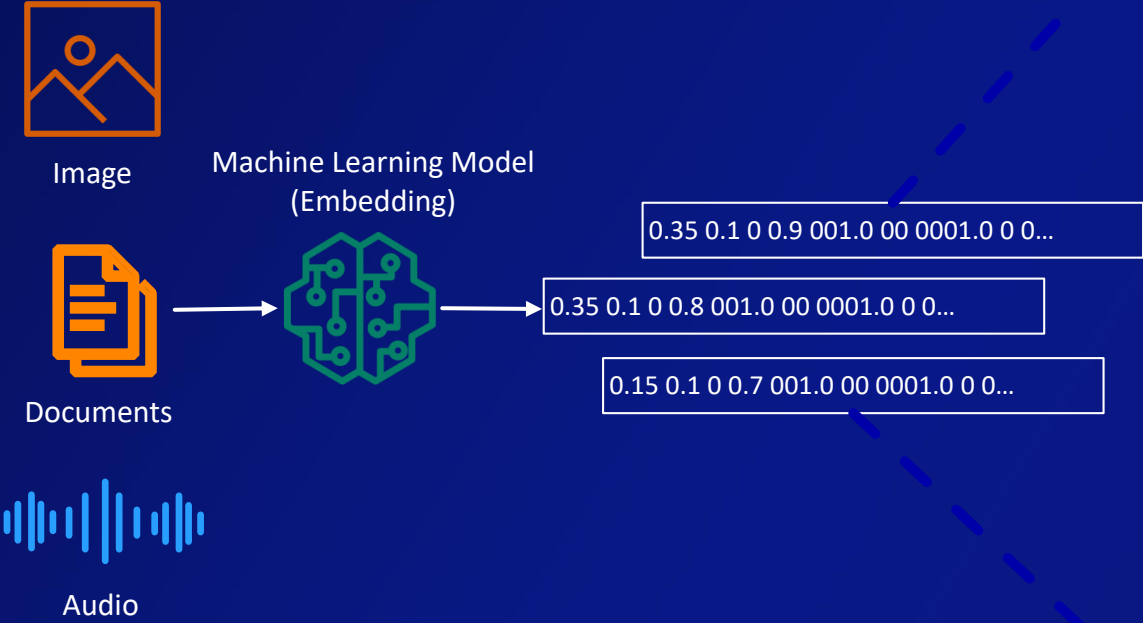
$$v_4 = [6, 4, 500, \$2.5M]$$

# What is a vector database?

## STORE AND RETRIEVE VECTOR EMBEDDINGS

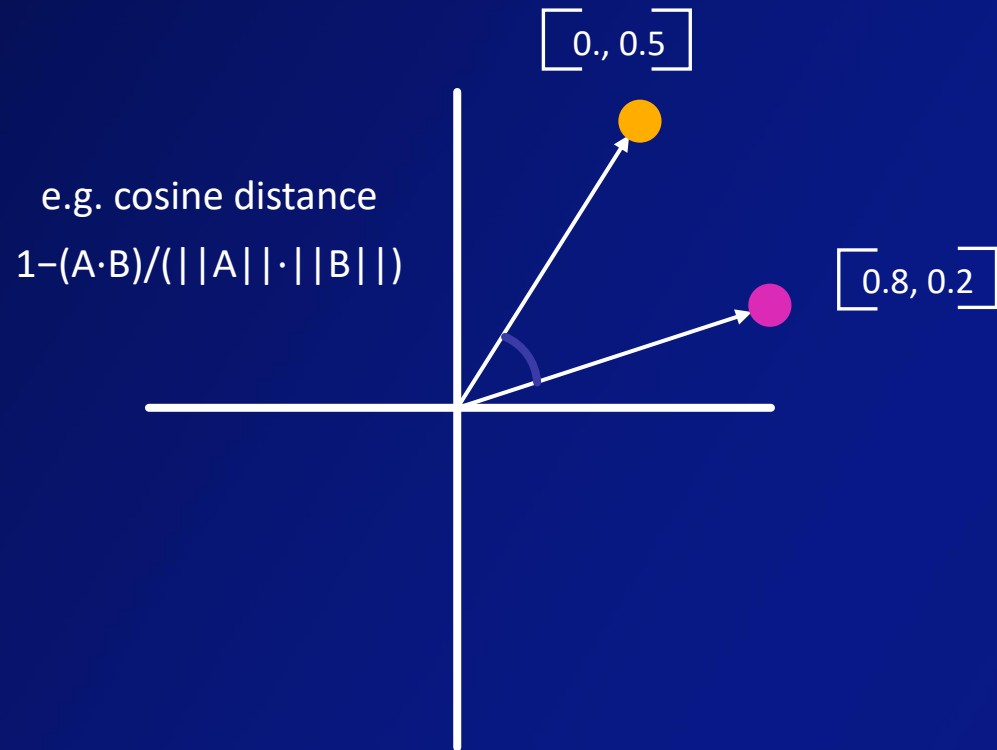
- A *vector datastore* or database enables you to store and query high dimensional vectors at scale, with efficient nearest neighbor query algorithms and appropriate indexes to improve data retrieval
- Enables updating and deleting vector embeddings as information changes e.g. product description updates
- Vector databases offer ability to filter information based on similarity and metadata e.g. embeddings of product description field, numeric fields for product price

# Data as Vectors





# Content (Vector) Similarity





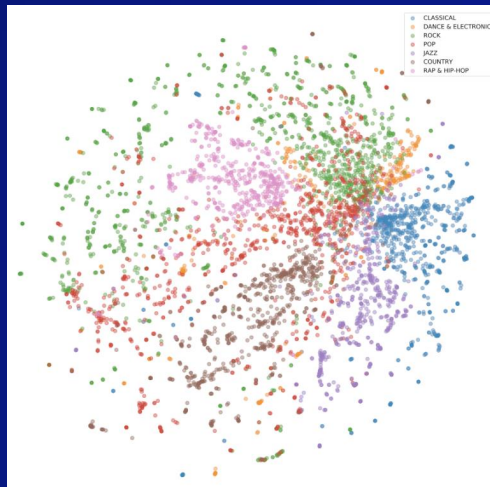
# k-Nearest-Neighbors: Find the “Top K” most similar...

## Exact (Brute-force) k-NN

query vector:  $\begin{bmatrix} 0.1, 0.2, 0.6 \dots 0.3, 0.7 \end{bmatrix}$   
(e.g. your favorite song)

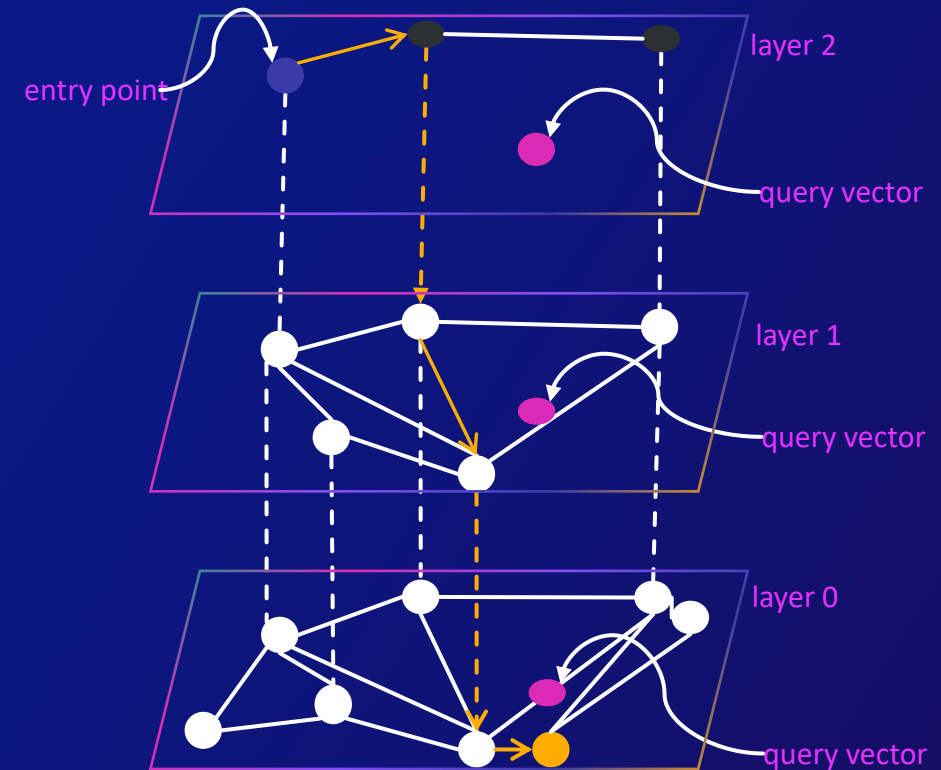
Calculate distance and rank  $\updownarrow$

Song  
vector  
corpus



## Approximated k-NN

Hierarchical Navigable Small Worlds (HNSW)



# Rich configurations, expert-driven process

Select Index Parameters

## Algorithms:

HNSW, ef\_construction, m...

## Quantization:

Scalar, Binary,  
Product

## Engine Settings:

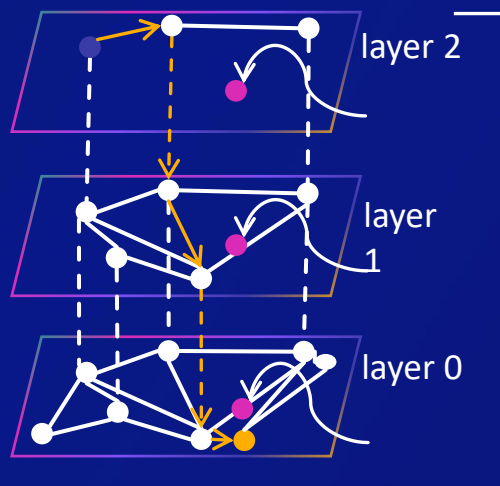
Disk-optimized,  
In-memory,  
Infrequent Queries

Build and Evaluate Index



1

Search latency



2

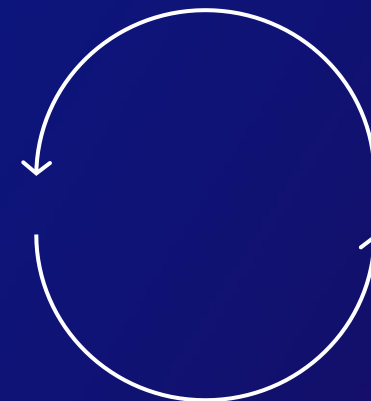
Memory  
Footprint

3

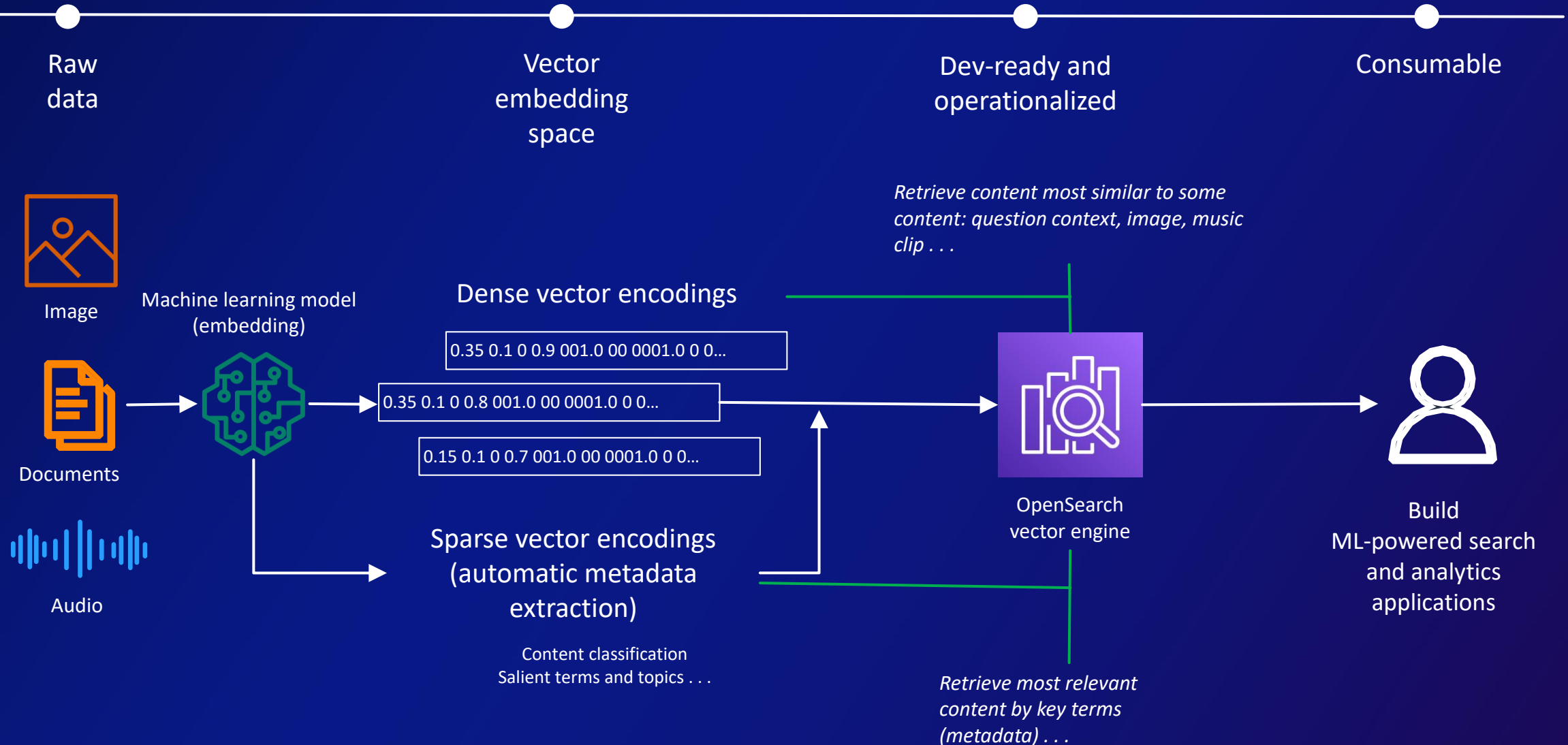
Recall

Exact (Brute-force) k-NN

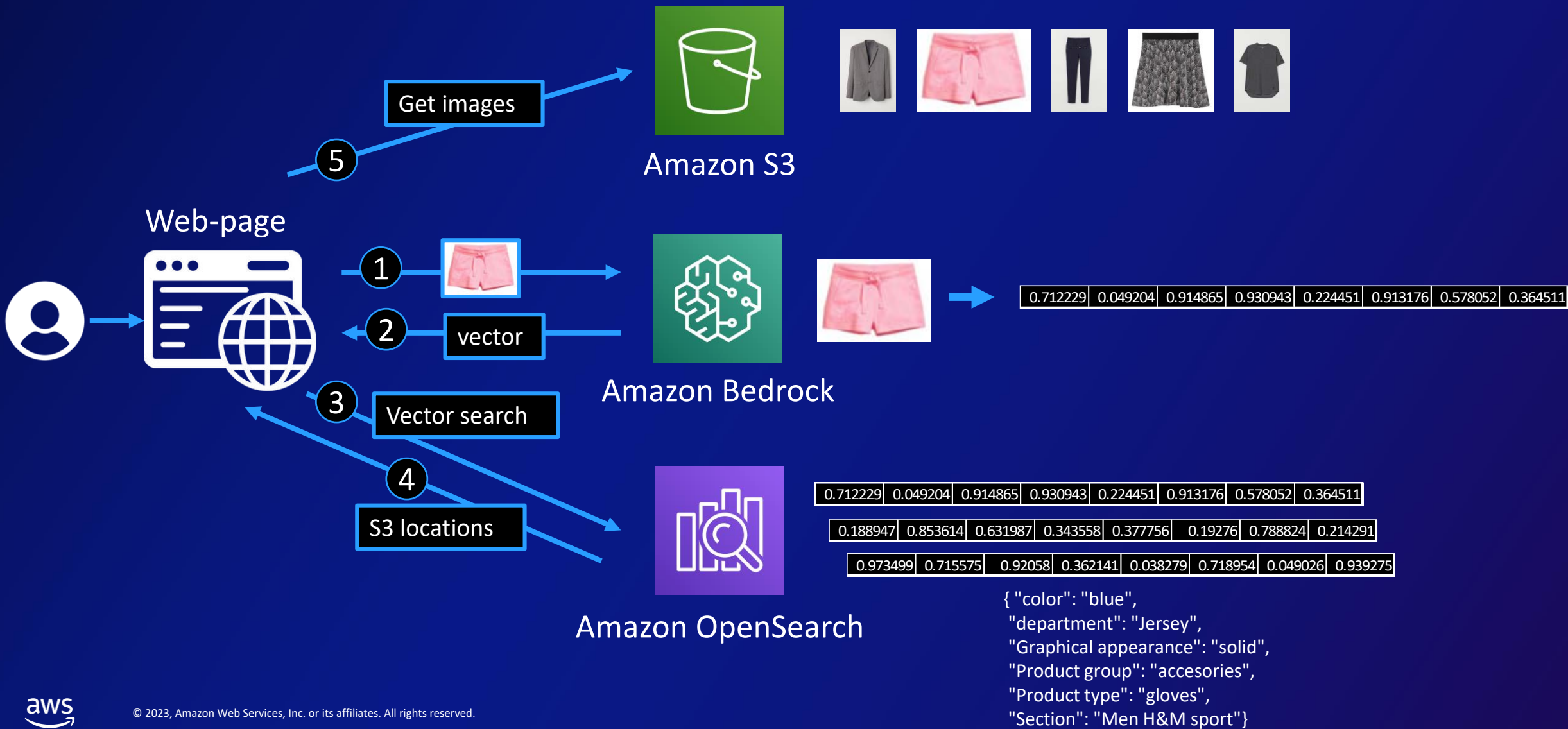
Adjust Parameters and Repeat



# What do I do with vectors?



# How Does AI Powered Search Work?



# Tiered vector storage

COST REDUCTIONS AT MASSIVE SCALE

**Exact KNN**

MOST ACCURATE & EXPENSIVE

**In-memory**

LOWER COST & HIGH PERFORMANCE

**Disk mode**

CHEAPER STORAGE, STILL PERFORMANT

**S3 vectors**

CHEAPEST, MASSIVE SCALE STORAGE

# Finding the Optimal trade-offs





# Amazon OpenSearch Serverless



## Easy to administer

No sizing, scaling, and tuning of clusters, and no shard and index lifecycle management



## Fast

Automatically scale resources to maintain consistently fast data ingestion rates and query response times



## Ecosystem

Get started in seconds using the same OpenSearch clients, pipelines, and APIs



## Cost-effective

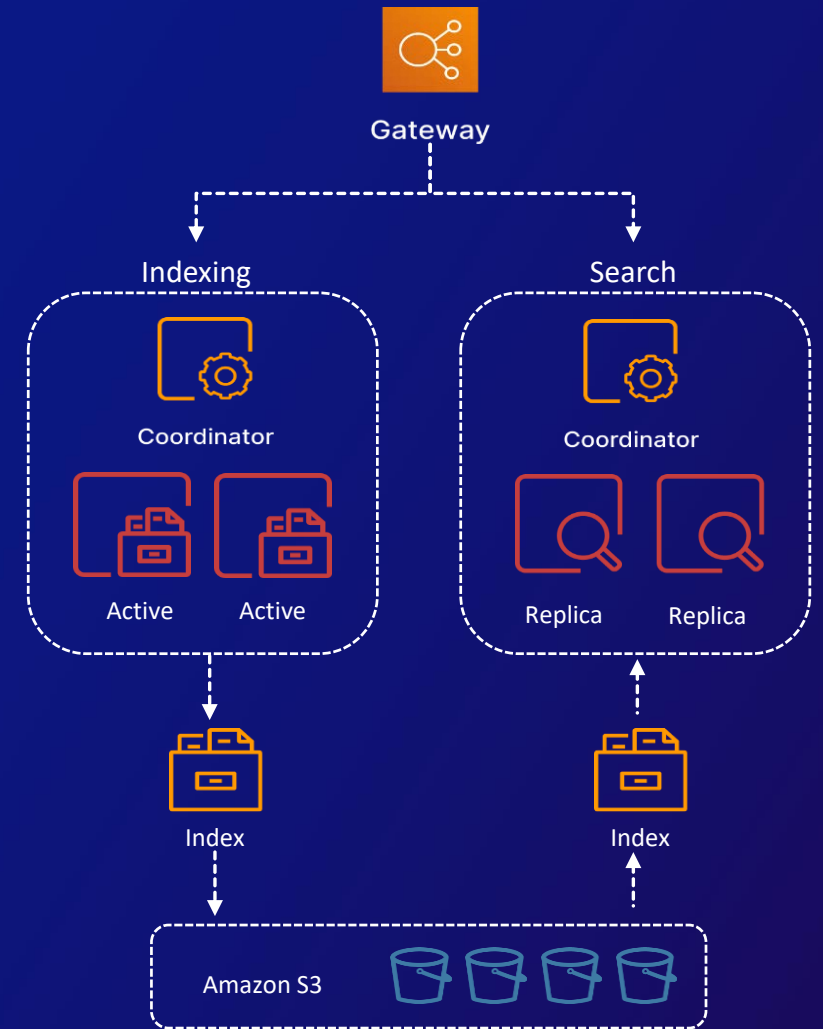
Pay only for the resources consumed



# OpenSearch Serverless Innovation

## NEW CAPABILITIES

- 100TB TIME-SERIES COLLECTIONS
- REGION EXPANSION FROM 15 TO 22 REGIONS
- DATA PLANE AUDIT LOGGING
- SNAPSHOT RESTORE



OpenSearch Serverless Modular Architecture

# Build Production Ready AI-powered Search Application

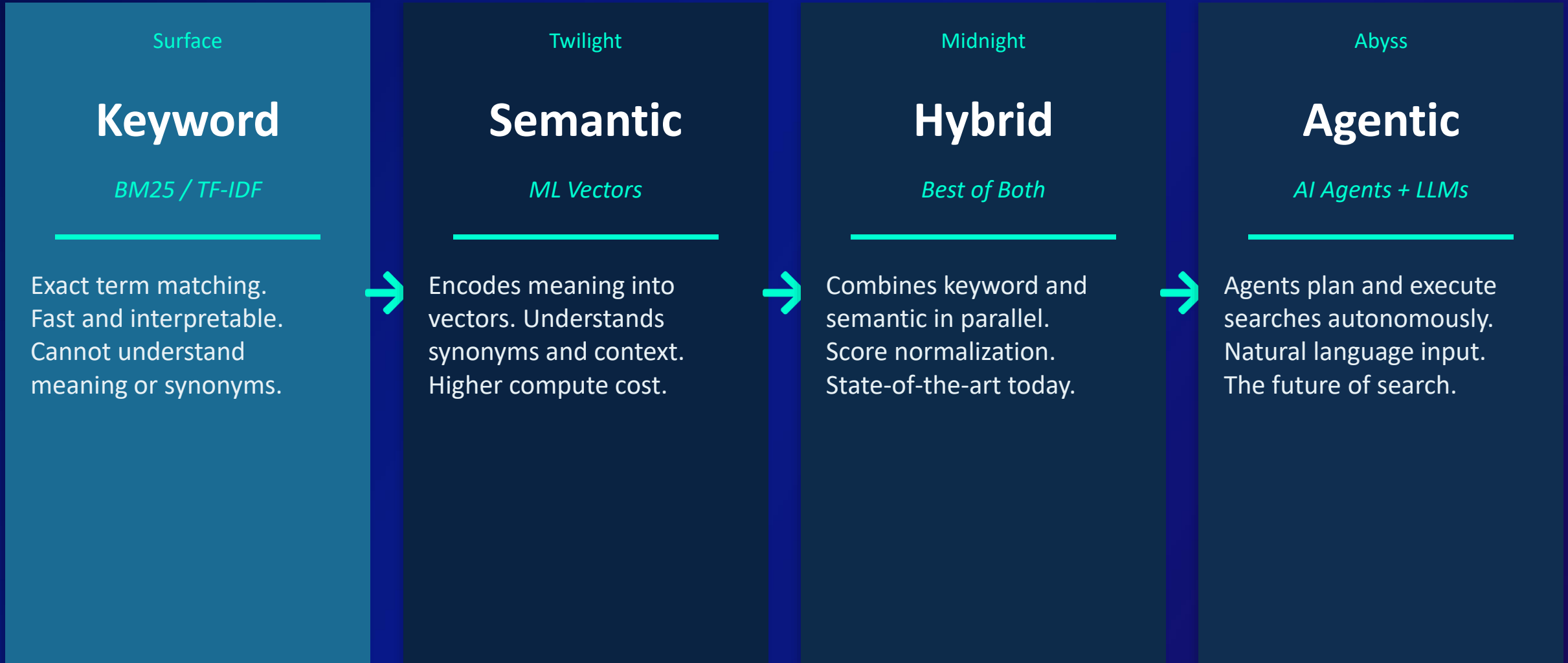
Prashant Agrawal

(He/Him)

Principal Solutions Architect

AWS

# The Evolution of Search



# Semantic Search — Two Approaches

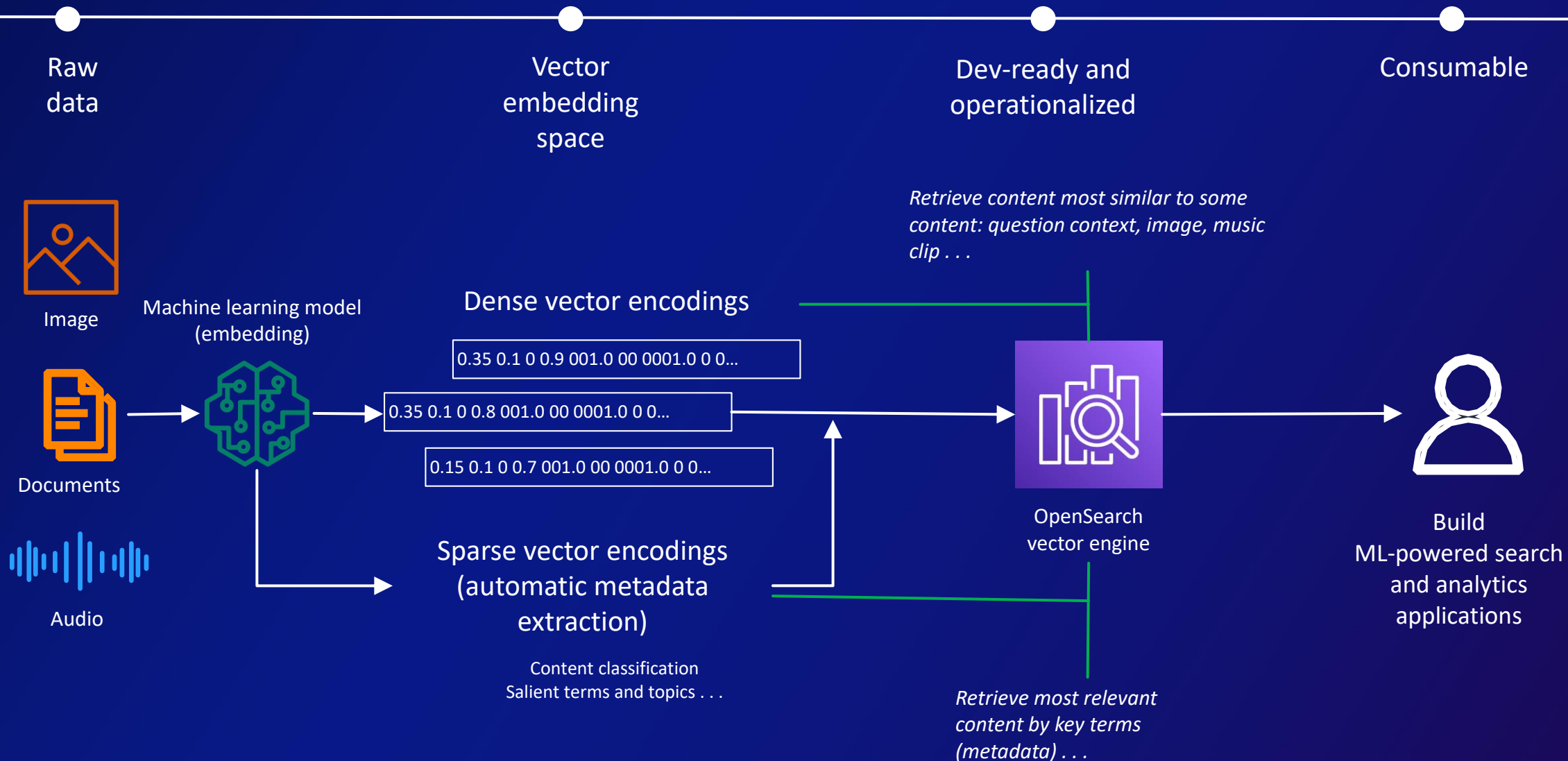
## Sparse Encoding

- Expands documents with weighted synonyms
- Uses existing inverted index (Lucene-native)
- 30,522 dims (BERT vocab) — only ~1% non-zero per doc
- 105,879 dims for multilingual variant
- As efficient as BM25 at query time
- Pre-trained: opensearch-neural-sparse-v2

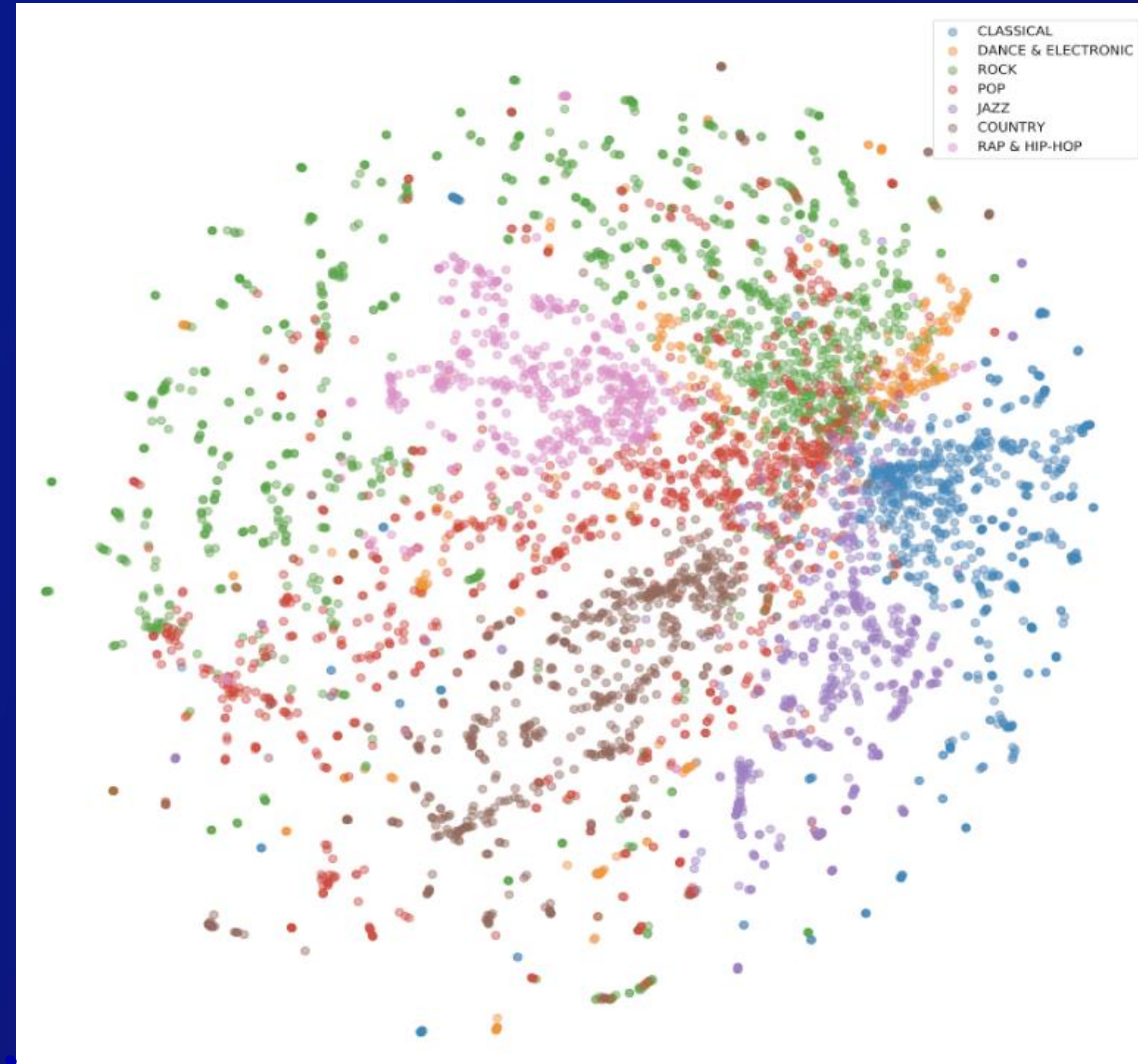
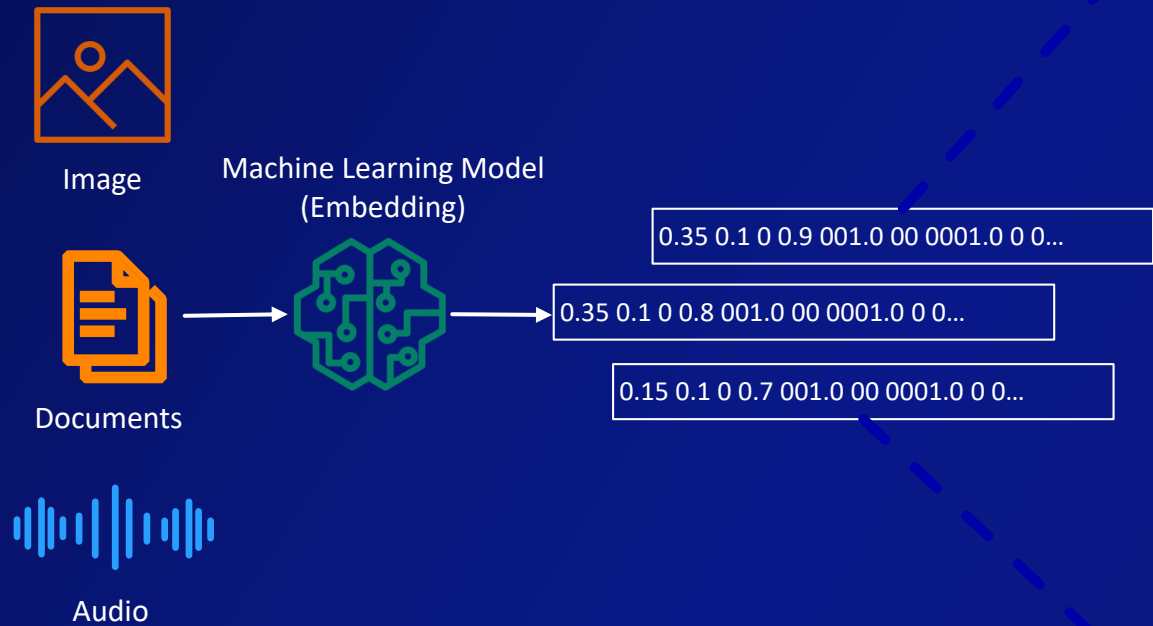
## Dense Encoding

- Fixed-length continuous vectors (384-768 dim)
- Stored in vector index (HNSW, IVF, etc.)
- Deep semantic relationships in vector space
- k-NN search at query time
- BERT, Cohere, OpenAI, Amazon Titan, Bedrock

# What do I do with vectors?



# Data as Vectors

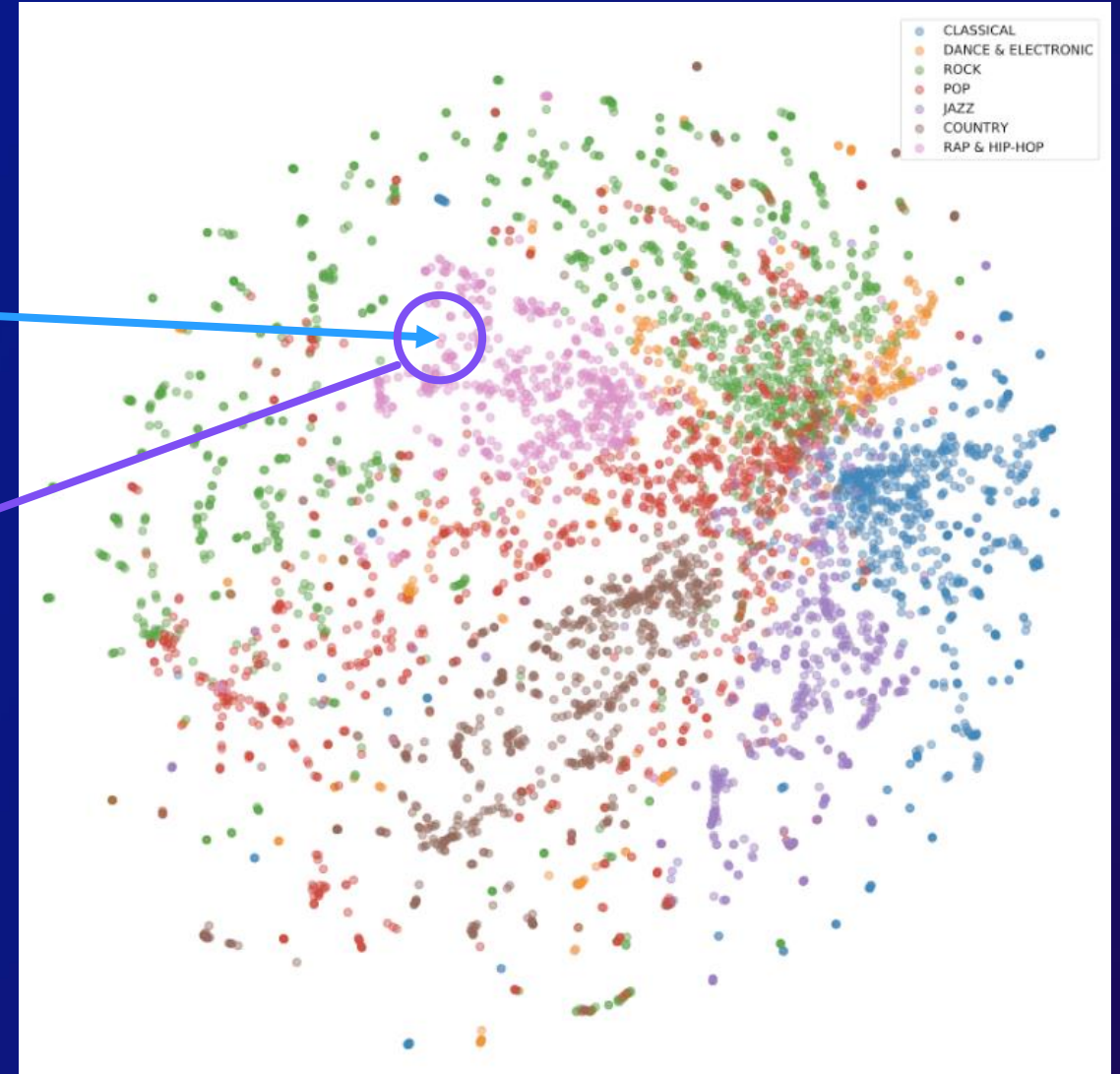


# Amazon Music : Recommendations

If you loved this song

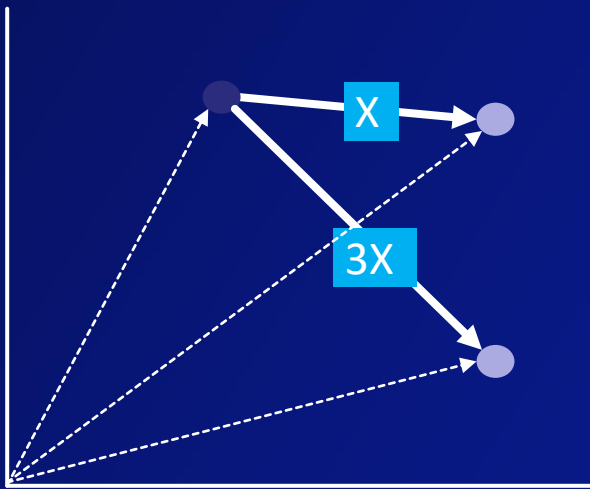
High chance you would like these 5 songs:

...  
...  
...

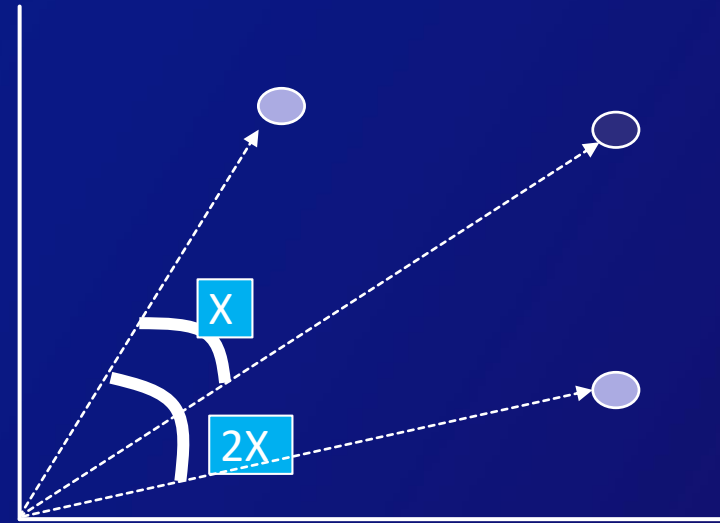




# Scoring semantic search



**Euclidean** or Straight line



**Cosine** = Angle between vectors

# Dense Encoding — Full Power of Embeddings



## Vector Space

Text converted to fixed-length vectors.  
Similarity via cosine, dot product, L2.  
Captures deep semantics and nuance.



## AI-Ready

Powers RAG pipelines with LLMs.  
Cross-lingual search capability.  
Works with BERT, Cohere, Titan.



## Optimized

HNSW, IVF, IVFPQ engines.  
FP16, INT8, binary quantization.  
50-97% memory reduction.

# Sparse similarity

Document 1: “**Apple** Products are expensive”

```
{ 'apple': 3.09,
  'expensive': 2.58,
  'apples': 2.31,
  'cost': 2.06,
  'products': 1.78,
  'cheap': 1.61,
  'product': 1.57,
  'price': 1.28,
  'expense': 1.04,
  'best': 0.66,
  'brand': 0.46,
  'stock': 0.44,
  'chip': 0.44,
  'store': 0.25,
  'computer': 0.24,
  'offer': 0.22,
  'money': 0.21,
  'budget': 0.21,
  'good': 0.2,
  'buy': 0.19,
  'affordable': 0.19,
  'popular': 0.16,
  'gift': 0.14,
  'manufacturer': 0.12,
  'purchase': 0.09,
  'iphone': 0.09,
  'happiness': 0.06,
  'steve': 0.03,
  'amazon': 0.03,
  'hardware': 0.01}
```

Document 2: “An **apple** a day keeps doctor away”

```
{ 'apple': 2.45,
  'doctor': 2.26,
  'day': 2.08,
  'away': 2.05,
  'apples': 1.87,
  'doctors': 1.86,
  'daily': 1.83,
  'keep': 1.59,
  'medical': 1.19,
  'prevent': 0.97,
  'every': 0.95,
  'a': 0.94,
  'fruit': 0.91,
  'dr': 0.7,
  'keeping': 0.62,
  'help': 0.56,
  'stay': 0.46,
  'an': 0.42,
  'remove': 0.41,
  'remedy': 0.37,
  'drink': 0.35,
  'one': 0.31,
  'candy': 0.31,
  'pill': 0.31,
  'keeps': 0.26,
  'diet': 0.24,
  'eat': 0.23,
```

Query: “**apple headphones**”

```
{ 'apple': 3.298149,
  'gift': 0.22470483,
  '##phone': 1.6386933,
  'electronics': 0.18978065,
  '##phones': 2.0765076,
  'music': 0.725811,
  'dj': 0.36222592,
  'sound': 0.77559084}
```

$score(D1, Q) = 7.4521604 > score(D2, Q) = 5.5605326$

Scoring:

$$score(D, Q) = \sum w_j \cdot w_i$$

# Sparse Encoding — Fast, Cheap, and Smart

10x

Smaller index than  
dense encoding

0

RAM increase at  
search time

- Leverages existing Lucene inverted index infrastructure
- Creates weighted synonyms: "car" → automobile: 0.8, vehicle: 0.6
- Graceful fallback to keyword matching — never worse than BM25
- Fully interpretable via the explain API
- Document-only mode eliminates online inference at query time
- Two-phase search acceleration (OpenSearch 2.15+)

# Why One Method May Not Always Work

## When Keyword Search Fails

- "Best laptop for students" — no exact match for "best"
- "Heart attack" vs "myocardial infarction"
- Conceptual: "things to do when bored"
- "Python" — the snake or the language?

## When Semantic Search Fails

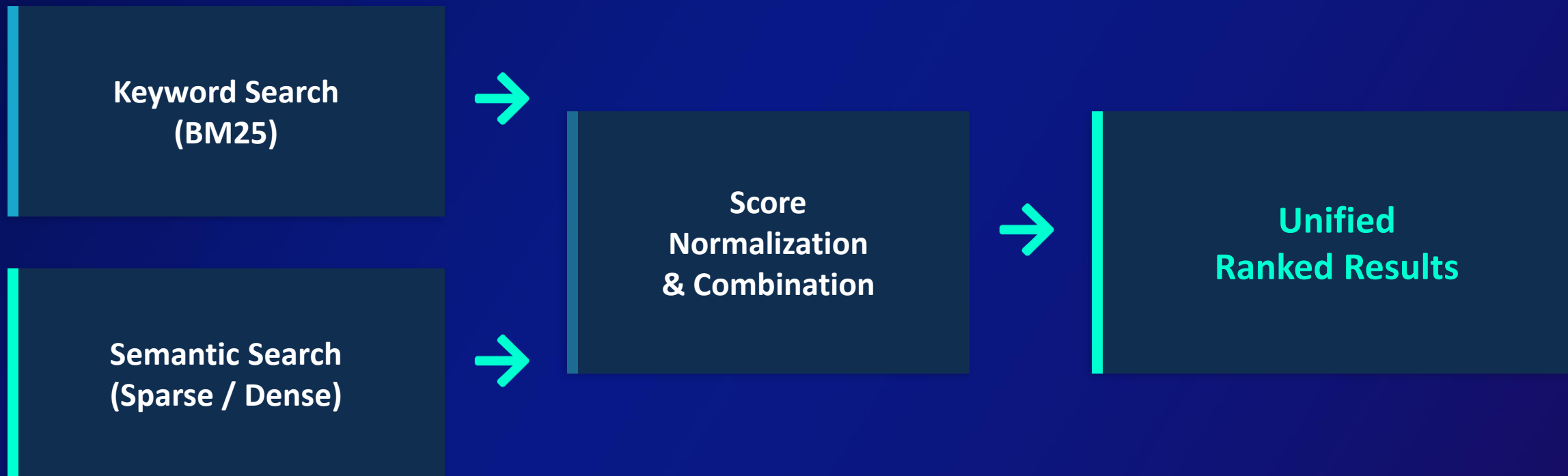
- Product SKUs, error codes: ERR-4032
- Domain-specific jargon without fine-tuning
- Precision-critical legal/compliance queries
- Numerical and date filtering needs



## The Takeaway

No single method covers all scenarios. The best search systems combine approaches intelligently. Users just want the right answer.

# Hybrid Search — Best of Both Worlds



## Arithmetic Mean

Equal weight to both

## Weighted Sum

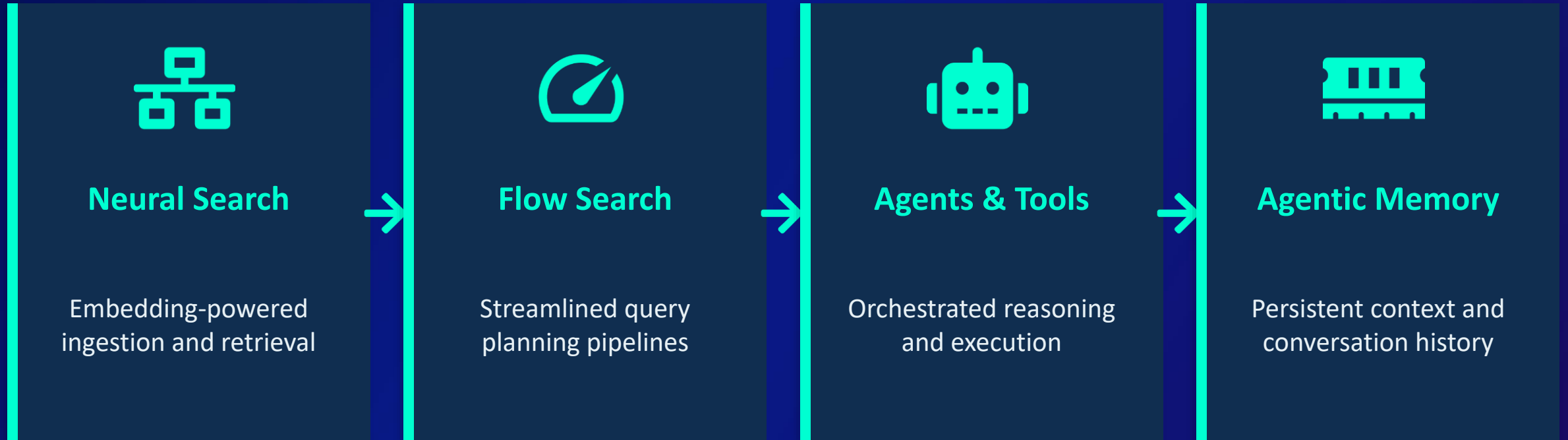
Tunable: 0.6 semantic + 0.4 keyword

## Harmonic Mean

Penalizes large discrepancies

# Agentic Search — When Search Starts Thinking

*Available in OpenSearch 3.3+ | Natural language queries | Transparent reasoning*



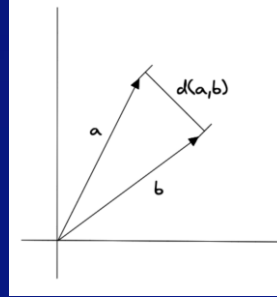
Built on OpenSearch ML Commons framework — agents, tools, models, and memory. Enabled through a simple 'agentic' query clause.



# KNN Distance Metrics

## Euclidean (L2)

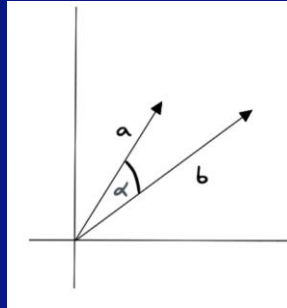
Useful for counts / measurements  
Recommendation Systems



$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

## Cosine Similarity

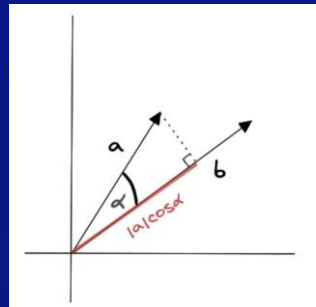
Useful for semantic search and document  
classification



$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}|| \cdot ||\mathbf{b}||}$$

## Dot Product

Useful for collaborative filtering

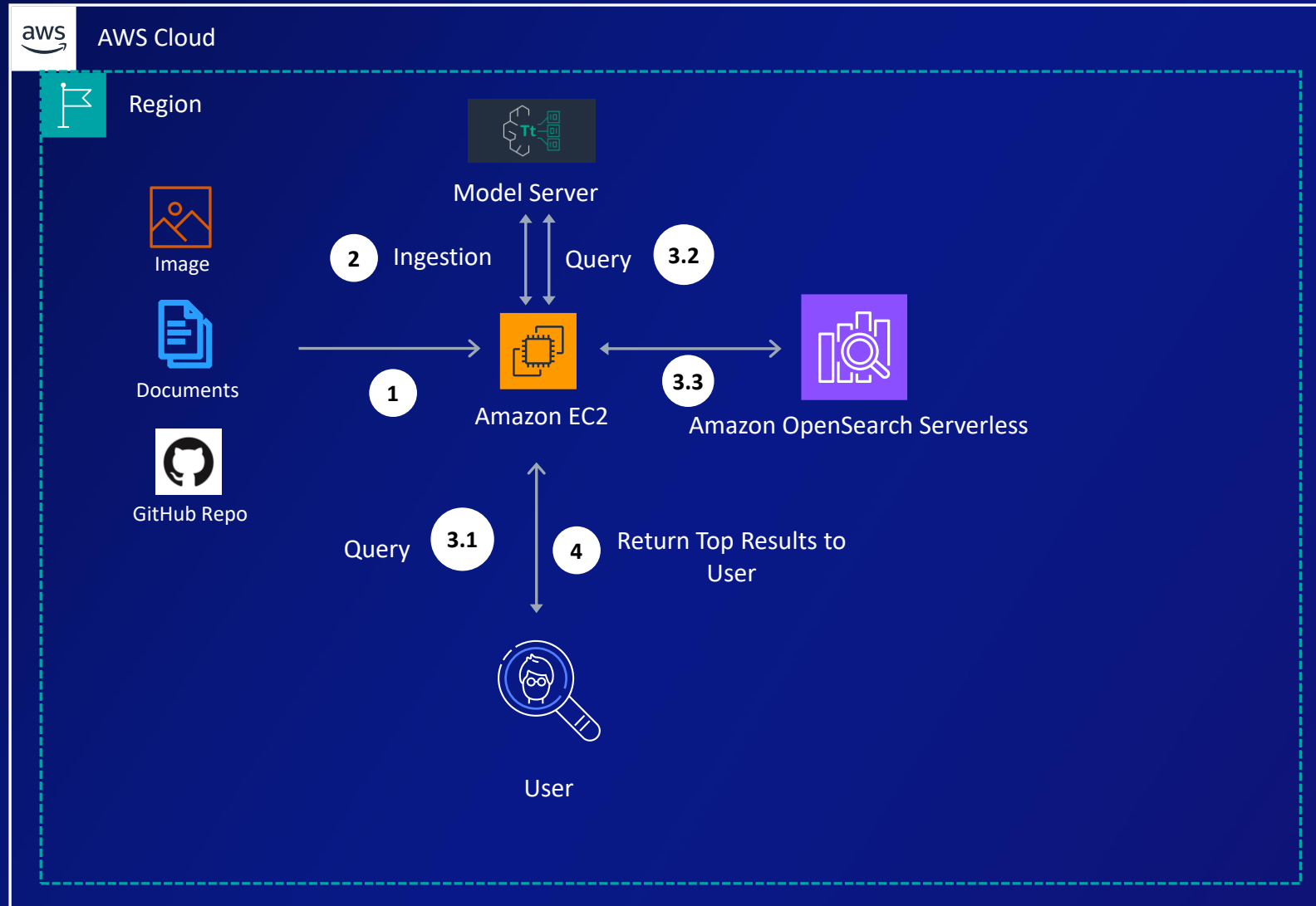


$$\mathbf{a} \cdot \mathbf{b} = ||\mathbf{a}|| ||\mathbf{b}|| \cos \alpha$$

# Architecture deep-dive

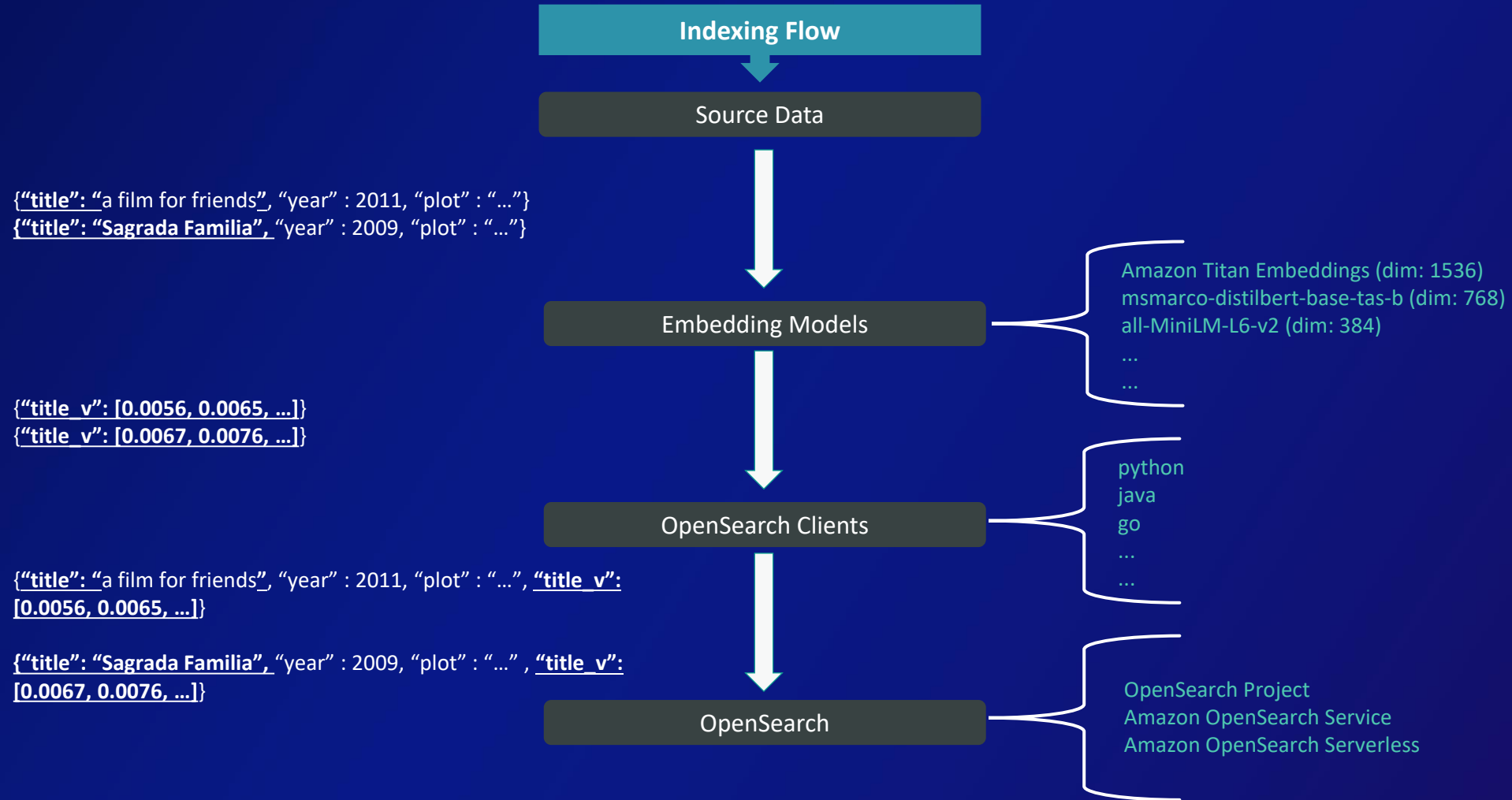


# AI-powered Search - Architecture



- 1 Use Amazon EC2 to crawl GitHub repo / other document sources and load Embedding model over there
- 2 Generate embeddings and ingest into OpenSearch Serverless index
- 3 Send vectorized query text for similarity search using k-NN plug-in
- 4 Relevant results are returned to user.

# AI-powered Search - Ingestion flow



# AI-powered Search - Ingestion (code snippet)

## Define Model and Generate embedding

# Load model (creates vector with 1536 dimensions)

```
bedrock_runtime.invoke_model(  
    modelId='amazon.titan-embed-text-v1',  
    contentType='application/json',  
    accept='application/json',  
    body=json.dumps({  
        'inputText': text  
    })  
)
```

# Retrieve Embedding

```
response_body = json.loads(response['body'].read())  
embeddings=response_body['embedding']
```

# Prepare Doc

```
vector_document = {  
    "title": title,  
    "plot": plot,  
    "v_title": embeddings,  
    ...  
}
```

## Setup OpenSearch Client

# Setup Credentials

```
credentials = boto3.Session().get_credentials()  
awsauth = AWS4Auth(credentials.access_key, credentials.secret_key,  
    region, service,  
    session_token=credentials.token)
```

# Create an OpenSearch client

```
client = OpenSearch(  
    hosts = [{'host': host, 'port': 443}],  
    http_auth = awsauth,  
    timeout = 300,  
    use_ssl = True,  
    verify_certs = True,  
    connection_class = RequestsHttpConnection  
)
```

# Index Documents

```
client.indices.create(  
    index=index_name,  
    body=index_body  
)
```

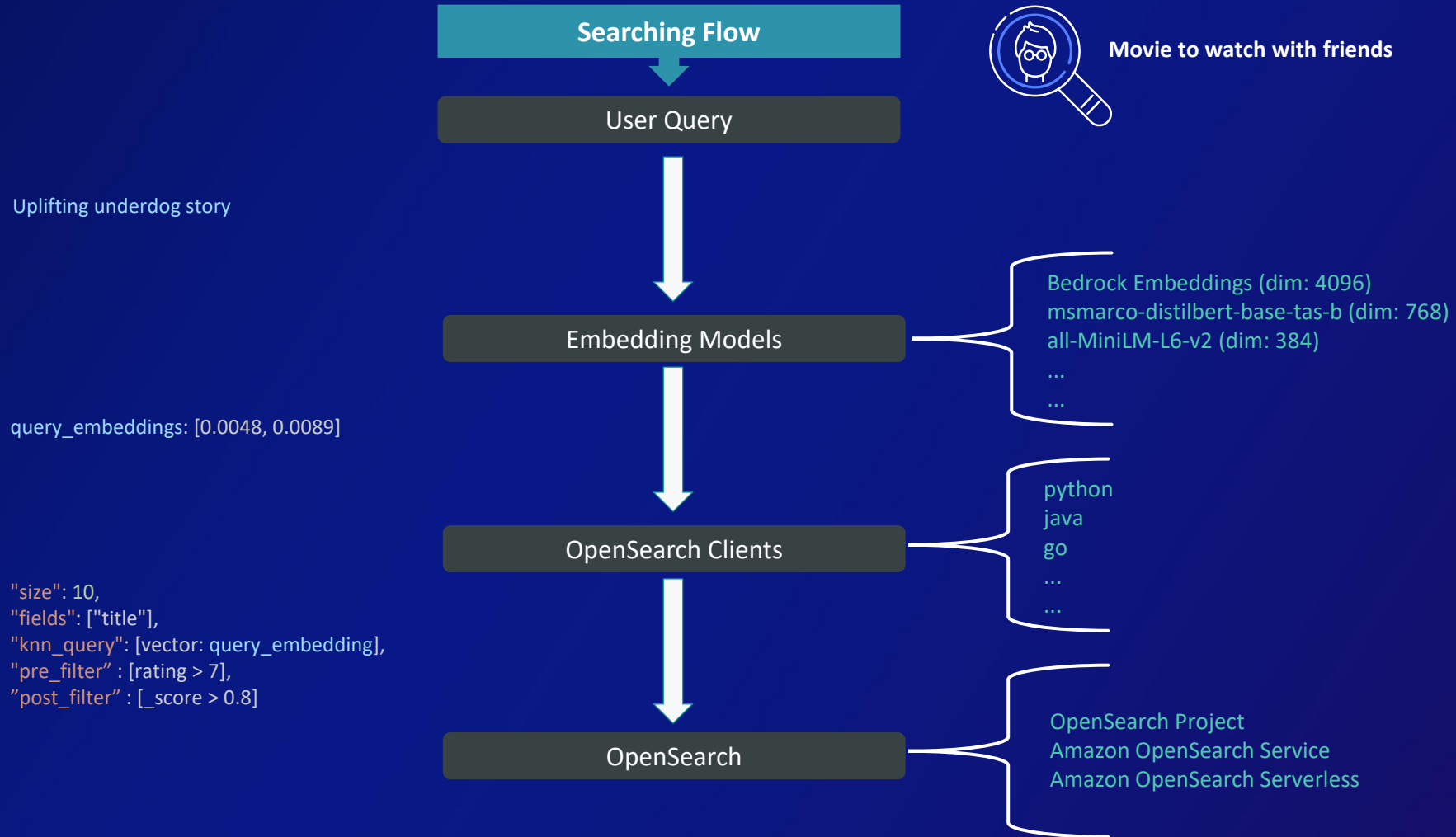
## Load data to OpenSearch

# create a new index

```
index_body = {  
    "settings": {  
        "index.knn": true  
    },  
    "mappings": {  
        "properties": {  
            "v_title": {  
                "type": "knn_vector",  
                "dimension": 1536  
            },  
            "title": { "type": "text" },  
            "plot": { "type": "text" },  
            "repository": { "type": "text" }  
        }  
    }  
}
```

# Bulk Index to OpenSearch Index  
client.bulk(body=actions)

# AI-powered Search - Search flow



# AI-powered Search - Search (code snippet)

## Use defined Model to Generate embedding for query

# Load model (creates vector with 1536 dimensions)

```
bedrock_runtime.invoke_model(  
    modelId='amazon.titan-embed-text-v1',  
    contentType='application/json',  
    accept='application/json',  
    body=json.dumps({  
        'inputText': text  
    })
```

# Retrieve Embedding

```
response_body = json.loads(response['body'].read())  
query_embedding=response_body['embedding']
```

# Prepare Query

```
"size": 3,  
"fields": ["title"],  
"knn_query": [vector: query_embedding],  
"pre_filter" : [rating > 7],  
"post_filter" : [_score > 0.8]
```

## Prepare OpenSearch Query

# Prepare OpenSearch Query

```
query_knn = {  
    "size": 3,  
    "_source": { "includes": [ "title" ..},  
    "query": {  
        "bool": {  
            "should": [  
                {  
                    "knn": {  
                        "v_title": {  
                            "vector": query_embedding,  
                            "k": 5  
                        }  
                    }  
                },  
            ],  
            "filter": [  
                {  
                    "range": {  
                        "rating": {  
                            "gte": 8  
                        }  
                    }  
                }  
            ]  
        }  
    }  
}
```

## Query OpenSearch

```
response = client.search(  
    body = query_knn,  
    index = index  
)
```

# Extract relevant information from the search result  
hits\_knn = response\_knn['hits']['hits']

```
doc_count_knn = response_knn['hits']['total']['value']
```

# Let's Build



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# Demo

Home

Semantic Search

Search Filters

Sort By

score

Select Genre

\*

Enter rating

5.00

0.0010.00

Compare lexical search with semantic search

This demo shows the difference between lexical search and semantic search for same queries across sample data sets for Movies, Product Q&A of Amazon products etc

Enter your search term

uplifting underdog story

Semantic Search using kNN

Showing 3 out of 565 matched documents

Million Dollar Baby (2004)

A determined woman works with a hardened boxing trainer to become a professional.

8.1 🌟 Drama, Sport



Million Dollar Baby

Brief Encounter (1945)

Meeting a stranger in a railway station, a woman is tempted to cheat on her husband.

8.1 🌟 Drama, Romance



Brief Encounter

Lexical Search using keywords

Showing 3 out of 42 matched documents

Sultan (2016)

Sultan is a classic underdog tale about a wrestler's journey, looking for a comeback by defeating all odds. Will he emerge victorious?

7.2 🌟 Action, Drama, Family



Sultan

Toy Story (1995)

A cowboy doll is profoundly threatened and jealous when a new spaceman figure supplants him as top toy in a boy's room.

8.3 🌟 Animation, Adventure, Comedy



Toy Story

aws

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# Workshop Felicitators



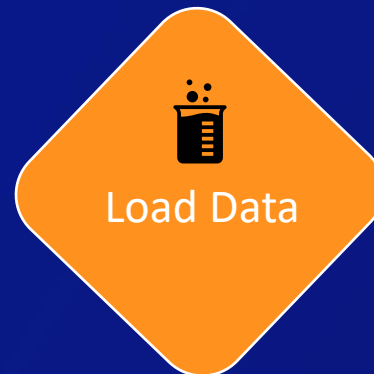
# Workshop Modules



Build an AI-powered semantic search application using Amazon OpenSearch Serverless and Amazon Bedrock



Update Security Group rules (ports 22 + 8080), connect to EC2 via Instance Connect, verify environment variables



Transform & Index: Run `movies_loader.py` to generate Bedrock Titan embeddings and bulk-index movie data into OpenSearch Serverless



Run `vector_search_bedrock.py` for interactive k-NN queries, then launch Streamlit app to compare keyword vs semantic

## Before you start



You have access to an AWS account with everything needed to complete this workshop.



The AWS account is only available for the duration of this workshop. **You will lose access to the account once the workshop is complete.**



The workshop is deployed to a specific AWS Region. Make sure that you are working in this Region as other Regions are blocked.



Review the terms and conditions of the event. **Do not upload any personal or confidential information to the account.**

## Accessing the workshop

1.

Sign in using your preferred method

<https://tinyurl.com/ai-search-513>

2.

Enter the One-time email passcode

**XXXXXXXXXX**

3.

Review terms and join event

4.

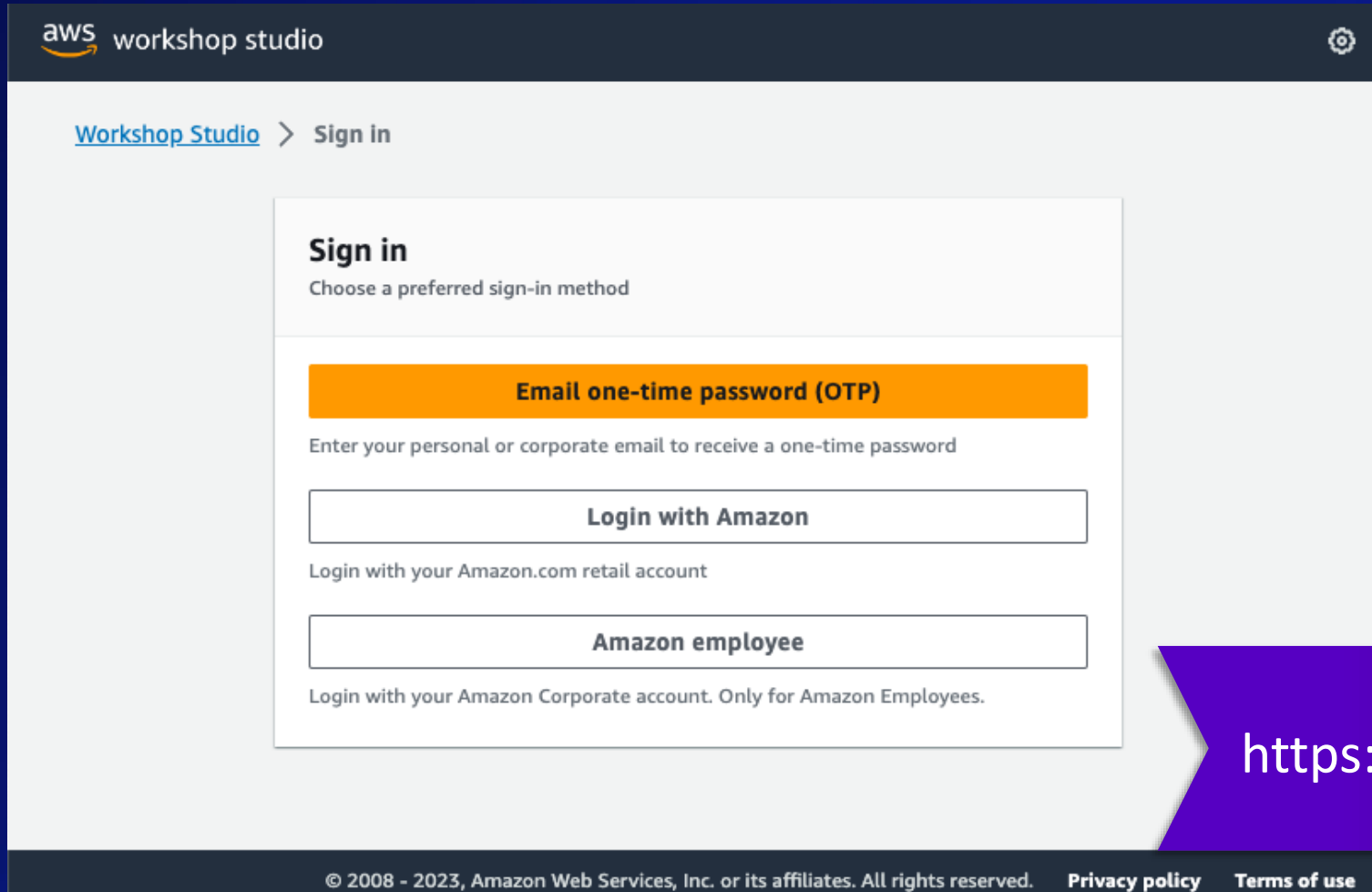
Click “Get Started” to begin the workshop

5.

Access AWS Accounts

Access the AWS Management Console

# Step 1: Sign in using your preferred method



The screenshot shows the AWS Workshop Studio sign-in interface. At the top, the 'aws workshop studio' header is visible. Below it, a breadcrumb trail shows 'Workshop Studio' followed by a chevron and 'Sign in'. The main content area is titled 'Sign in' with the instruction 'Choose a preferred sign-in method'. There are three sign-in options: 1. 'Email one-time password (OTP)' in an orange box, with the instruction 'Enter your personal or corporate email to receive a one-time password'. 2. 'Login with Amazon' in a white box, with the instruction 'Login with your Amazon.com retail account'. 3. 'Amazon employee' in a white box, with the instruction 'Login with your Amazon Corporate account. Only for Amazon Employees.' The footer contains the copyright notice '© 2008 - 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.' and links for 'Privacy policy' and 'Terms of use'.

aws workshop studio

[Workshop Studio](#) > Sign in

## Sign in

Choose a preferred sign-in method

**Email one-time password (OTP)**

Enter your personal or corporate email to receive a one-time password

**Login with Amazon**

Login with your Amazon.com retail account

**Amazon employee**

Login with your Amazon Corporate account. Only for Amazon Employees.

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<https://tinyurl.com/ai-search-513>

# Step 2: Enter the One-time email passcode

## One-time email passcode

We sent a passcode to prashanttct07@gmail.com. You should receive it within 5 minutes.

Passcode (9-digit)

Resend passcode

Code expires in 14:52

Back

Sign in

[Get help signing in](#)

# Step 3: Review terms and join event

aws workshop studio

Workshop Studio > Join event

Step 1  
[Enter event access code](#)

Step 2  
**Review and join**

Review and join

Event details

|                            |                    |          |       |
|----------------------------|--------------------|----------|-------|
| Name                       | Start time         | Duration | Level |
| re:Invent 2023 Workshop    | 8/28/2023 10:59 AM | 12 hours | 100   |
| Description                |                    |          |       |
| Workshop at re:Invent 2023 |                    |          |       |

Terms and Conditions

Read and accept before joining the event

Read and accept before joining the event:

1. By using AWS Workshop Studio for the relevant event, You agree to [the AWS Event Terms and Conditions](#) and the [AWS Acceptable Use Policy](#).

2. If You are under 18 years old, You may participate in the relevant event, using AWS Workshop Studio: (a) if you are at least the minimum age below based on the country or region in which you reside, and (b) with the involvement of a parent, guardian or educator.

| Country or region                                                                                                                                                  | Minimum age |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| All countries or regions not listed below (including United States, Brazil, UK, and India)                                                                         | 13          |
| China, Republic of Korea (South Korea)                                                                                                                             | 14          |
| Australia                                                                                                                                                          | 15          |
| Japan, Switzerland, Vietnam, and countries in the European Economic Area                                                                                           | 16          |
| Azerbaijan, Bolivia, Colombia, Dominican Republic, Indonesia, Lebanon, Malaysia, Mexico, Montenegro, Nepal, Philippines, Thailand, Turkey, and countries in Africa | 18          |

3. You acknowledge and agree that You are using an AWS-owned account that You will be only be able to access during the relevant event. You have no ownership rights over this AWS-owned account.

4. During the relevant event, while using this AWS-owned account, You will not use, import, input, or introduce any data, dataset, or other material that contains personal data, financial information, or any other data or materials that may be subject to laws and regulations (such as the General Data Protection Regulation or The Health Insurance Portability and Accountability Act of 1996).

5. If You find residual resources or materials in this AWS-owned account, You will notify Your Event Operator immediately.

6. AWS, its affiliates and any entities or persons acting on AWS's behalf, reserves the right to terminate this AWS-owned account and to delete its contents at any time, without any notice to You.

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8. You will not copy, import, export or otherwise create derivative works of materials provided by AWS for use outside of the event.

9. AWS, its affiliates and any entities or persons acting on AWS's behalf, has no obligation to enable the transmission of Your materials through AWS Workshop Studio, and may, in its discretion, edit, block, refuse to post, or remove Your materials at any time, without notice to You.

10. If You are an AWS Partner, using AWS Workshop Studio as part of Your participation in the AWS Partner Network Program, Your use of AWS Workshop Studio is governed by these terms, the AWS Partner Network Terms and Conditions, and Your applicable customer agreement with AWS.

11. Your use of AWS Workshop Studio will comply with these terms and all applicable laws. If You fail to comply with any of these terms, Your access to AWS Workshop Studio may be immediately terminated, without notice to You.

☒ I agree with the Terms and Conditions

Cancel

Previous

Join event

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# Step 4: Get started with the workshop

The screenshot shows the AWS Workshop Studio interface. The top header displays the AWS logo, the text 'workshop studio', a settings icon, and the user profile 'prashanttct07'. The sidebar on the left contains the workshop title 'AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock' and a list of steps: Introduction, Step 1: Set Up Your AI-Powered Search Environment, Step 2: Transform Data for AI-Powered Search, Step 3: Implement AI-Powered Search, and Summary and Cleanup. Below the steps is a section for 'AWS account access' with links to 'Open AWS console (us-west-2)' and 'Get AWS CLI credentials'. The main content area features a blue banner at the top stating 'Event ends in 1 day 11 hours 56 minutes.' Below this is the workshop title and a breadcrumb trail. The 'Event information' section includes a table with details on start time, duration, accessible regions, and description. The 'Workshop' section includes a table with details on title, complexity level, topics, and AWS services. A blue arrow points to the 'Get started' button in the top right corner of the workshop details section.

aws workshop studio

Event ends in 1 day 11 hours 56 minutes.

Event dashboard > AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock

## AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock

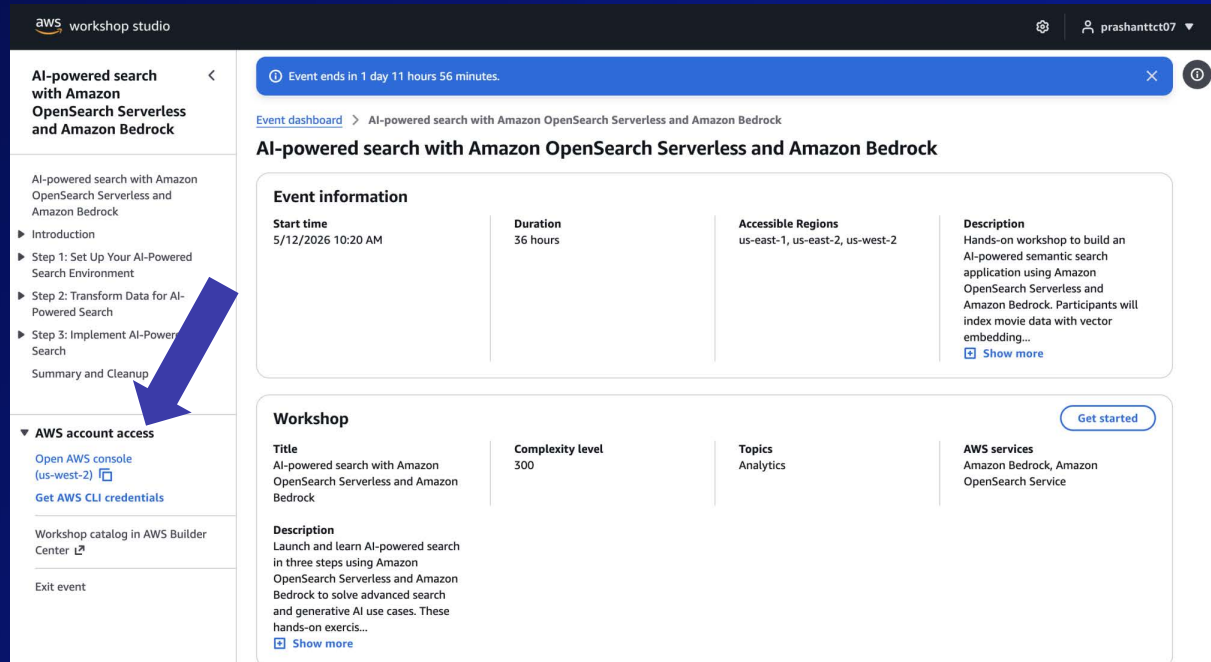
| Event information                       |                             |                                                              |                                                                                                                                                                                                                                            |
|-----------------------------------------|-----------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Start time</b><br>5/12/2026 10:20 AM | <b>Duration</b><br>36 hours | <b>Accessible Regions</b><br>us-east-1, us-east-2, us-west-2 | <b>Description</b><br>Hands-on workshop to build an AI-powered semantic search application using Amazon OpenSearch Serverless and Amazon Bedrock. Participants will index movie data with vector embedding...<br><a href="#">Show more</a> |

| Workshop                                                                                                                                                                                                                                   |                                |                            |                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------|------------------------------------------------------------------|
| <b>Title</b><br>AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock                                                                                                                                                     | <b>Complexity level</b><br>300 | <b>Topics</b><br>Analytics | <b>AWS services</b><br>Amazon Bedrock, Amazon OpenSearch Service |
| <b>Description</b><br>Launch and learn AI-powered search in three steps using Amazon OpenSearch Serverless and Amazon Bedrock to solve advanced search and generative AI use cases. These hands-on exercis...<br><a href="#">Show more</a> |                                |                            |                                                                  |

[Get started](#)



# Step 5: Access AWS Account



aws workshop studio

Event ends in 1 day 11 hours 56 minutes.

Event dashboard > AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock

### AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock

**Event information**

|                                         |                             |                                                              |                                                                                                                                                                                                                                            |
|-----------------------------------------|-----------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Start time</b><br>5/12/2026 10:20 AM | <b>Duration</b><br>36 hours | <b>Accessible Regions</b><br>us-east-1, us-east-2, us-west-2 | <b>Description</b><br>Hands-on workshop to build an AI-powered semantic search application using Amazon OpenSearch Serverless and Amazon Bedrock. Participants will index movie data with vector embedding...<br><a href="#">Show more</a> |
|-----------------------------------------|-----------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Workshop**

|                                                                                        |                                |                            |                                                                  |
|----------------------------------------------------------------------------------------|--------------------------------|----------------------------|------------------------------------------------------------------|
| <b>Title</b><br>AI-powered search with Amazon OpenSearch Serverless and Amazon Bedrock | <b>Complexity level</b><br>300 | <b>Topics</b><br>Analytics | <b>AWS services</b><br>Amazon Bedrock, Amazon OpenSearch Service |
|----------------------------------------------------------------------------------------|--------------------------------|----------------------------|------------------------------------------------------------------|

**Description**  
Launch and learn AI-powered search in three steps using Amazon OpenSearch Serverless and Amazon Bedrock to solve advanced search and generative AI use cases. These hands-on exercises...  
[Show more](#)

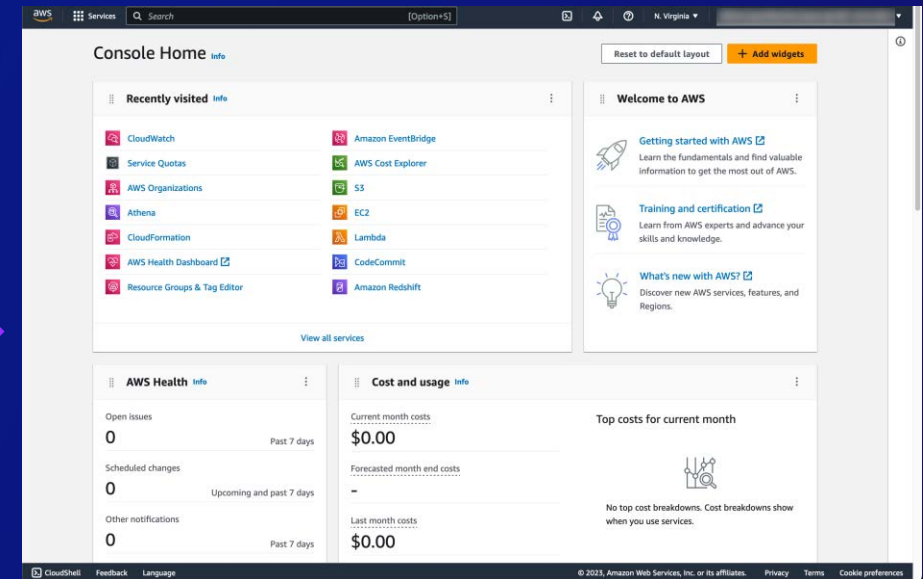
**Get started**

**AWS account access**

- [Open AWS console \(us-west-2\)](#)
- [Get AWS CLI credentials](#)

Workshop catalog in AWS Builder Center

Exit event



aws Services Search [Option+5] N. Virginia

### Console Home

Reset to default layout + Add widgets

**Recently visited**

- CloudWatch
- Service Quotas
- AWS Organizations
- Athena
- CloudFormation
- AWS Health Dashboard
- Resource Groups & Tag Editor
- Amazon EventBridge
- AWS Cost Explorer
- S3
- EC2
- Lambda
- CodeCommit
- Amazon Redshift

**Welcome to AWS**

- [Getting started with AWS](#)
- [Training and certification](#)
- [What's new with AWS?](#)

**AWS Health**

Open issues: 0 (Past 7 days)

Scheduled changes: 0 (Upcoming and past 7 days)

Other notifications: 0 (Past 7 days)

**Cost and usage**

Current month costs: \$0.00

Forecasted month end costs: -

Last month costs: \$0.00

Top costs for current month

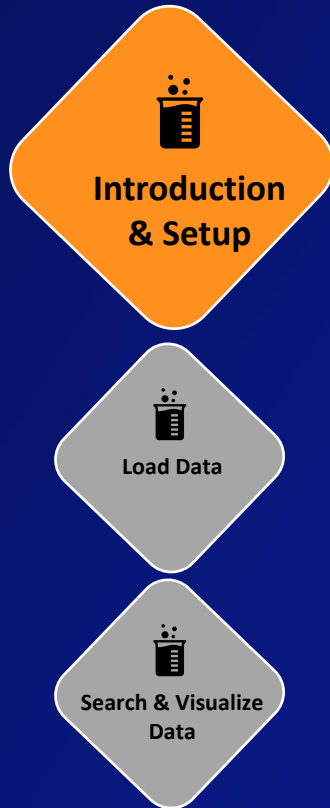
No top cost breakdowns. Cost breakdowns show when you use services.

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Access the AWS Management Console



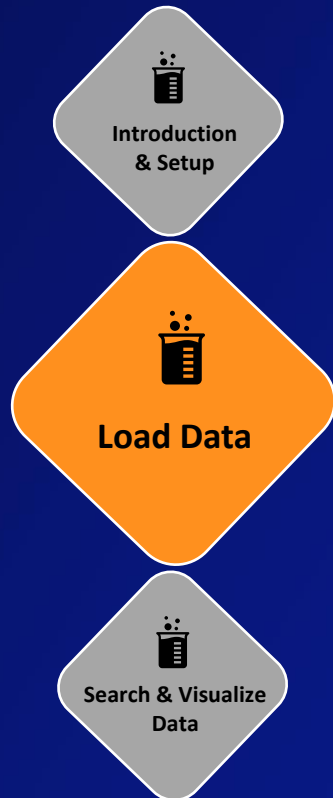
# Workshop Checkpoint - 1



By now you should have:

- ✓ Security group updated (ports 22 + 8080)
- ✓ Connected to EC2 via Instance Connect as ec2-user
- ✓ Run: `echo $AOSS_VECTORSEARCH_ENDPOINT` → shows your collection URL
- ✓ Run: `echo $AOSS_VECTORSEARCH_REGION` → shows us-west-2

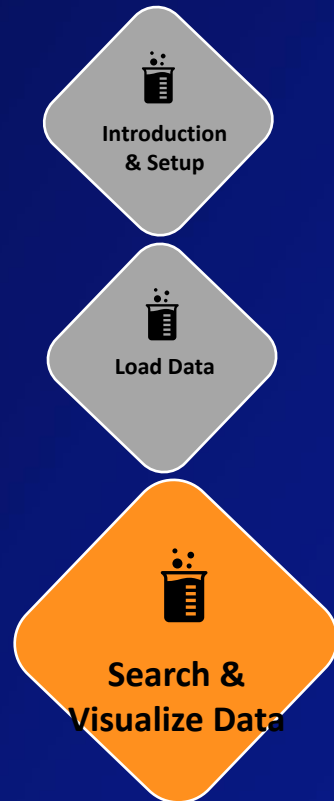
# Workshop Checkpoint - 2



By now you should have:

- ✓ `cd /home/ec2-user/vector-engine-demos/indexer`
- ✓ `python3 movies_loader.py --background` (NO sudo!)
- ✓ `tail -f movies_loader.log` → shows "bulk indexing done"

# Workshop Checkpoint - 3



By now you should have:

- ✓ `python3 vector_search_bedrock.py` → type query → see results
- ✓ `streamlit run 0_Home.py` → running on port 8080
- ✓ `http://<your-EC2-IP>:8080` → Streamlit app loads
- ✓ Semantic Search page → try a query!

# Best practices for cost optimization, performance tuning, and monitoring

# The Search System Design Triangle



# Cost Optimization — Strategies That Work

**50–97%**

memory savings  
with quantization

**10x**

smaller index with  
sparse encoding

**Quantization: FP32 → FP16 → INT8 → Binary**

Binary + rescoring = 97% memory reduction, <2% recall loss

**Disk-based vector search (OpenSearch 2.17+ )**

Vectors on SSD, not RAM — up to 32x memory reduction

**Sparse encoding (doc-only mode)**

Zero RAM overhead at query time, uses Lucene inverted index

**Right-size dimensions: 768 → 384**

50% storage, ~1% recall loss

**Tiered storage: Hot → Warm → Cold (S3)**

HNSW in-memory → Disk-based → S3 vectors

**Serverless OCU auto-scaling**

Pay only for actual search/indexing compute

# Performance Tuning—Latency Deep Dive

## HNSW Parameters

### **ef\_construction**

Index build quality. Higher = better graph, slower build.  
Only affects indexing time, not query performance.

### **M (graph connectivity)**

Higher M = better recall but more memory + slower indexing.  
Default 16 is good for most cases. Try 32 for high-recall.

### **ef\_search (query-time tuning)**

Start at 100, increase until recall plateaus.

**This is the knob most people undertune!**

## Query Optimization

### **Pre-filtering vs Post-filtering**

Pre-filter reduces candidate set → much faster search.  
Use when filter selectivity is high.

### **Approximate vs Exact**

ANN (HNSW) = milliseconds latency  
Exact k-NN = seconds latency

### **Request cache**

Cache repeated semantic queries for instant response.

### **Force-merge to 1 segment**

20–40% search latency reduction after bulk indexing.



# Recall Optimization — Getting Better Results

**+10–15%**

recall improvement  
with domain fine-tuning

**Hybrid**

BM25 + kNN consistently  
beats either alone

## **Hybrid search: BM25 + kNN in parallel**

Score normalization (min\_max vs L2) affects ranking quality

## **Reranking: Cross-encoder as second pass**

Cohere, BGE — improves precision without changing index

## **Chunking strategy**

10–20% overlap, 256–512 tokens optimal

Too large = diluted semantics, too small = lost context

## **Embedding model selection**

Domain fine-tuning on 5K examples > larger generic model

## **Multi-vector per document**

Title + Body + Summary embeddings separately

## **Metadata filtering: Narrow first, then semantic**

# Monitoring—What to Watch

## Quality & Relevance

### Recall@K

Are your top-K results actually relevant?  
Needs labeled data or click-through signals

### NDCG / MRR

Ranking quality — position matters, not just presence

### A/B testing

Compare approaches with real user traffic  
Click-through rate is the ultimate signal

### Cost per relevant result

Combines all three triangle dimensions  
Best single business metric

## Operational

### Latency P50 / P95 / P99

Track percentiles, not averages — tail matters  
P99 tells you about worst user experience

### Index size & segment count

Directly correlates with query latency  
Force-merge after bulk indexing

### OCU utilization (Serverless)

Over-provisioned or throttling?

### Embedding latency

Bedrock/model inference — often the real bottleneck

### Query volume patterns

Burst traffic → pre-warm indexes

# Putting It Together — Decision Framework

**Step 1:** Define your latency SLA ( $P99 < 200\text{ms?}$   $< 500\text{ms?}$   $< 2\text{s?}$ )

**Step 2:** Set minimum acceptable recall ( $\text{Recall}@10 > 0.85?$   $> 0.95?$ )

**Step 3:** Cost becomes the optimization variable

## E-commerce

Low latency + High recall  $\rightarrow$  Invest in infra  
In-memory HNSW, pre-filtering, aggressive caching

## Internal Knowledge Base

Moderate latency OK  $\rightarrow$  Optimize cost  
Disk-based vectors, sparse encoding, OCU auto-scale

## Legal / Compliance

Maximum recall required  $\rightarrow$  Accept higher cost + latency  
Exact k-NN, hybrid search, reranking pipeline

## Chatbot / RAG

Balance all three  $\rightarrow$  Hybrid search + caching  
Domain fine-tuned model, quantized vectors, tiered storage

# Any Questions?



# Thank you!

Prashant Agrawal  
prashagr@amazon.com

